Design and Validation of a Shaker Table with various motion modes

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Abstract

We applied the concepts of periodic motion and mechanical waves from the undergraduate physics curriculum to the design and development of a single-axis shaker table as a student project. The system consists of a table with 3D-printed components, a stepper motor, and an Arduino controller. Unlike conventional setups limited to sinusoidal oscillations, this shaker table can generate various motion modes to meet different user requirements. This paper demonstrates sinusoidal, trapezoidal and triangular wave motions. The performance of the system is tested and analyzed using an ultrasonic motion sensor, followed by an error analysis and a discussion of teaching practice.

I. INTRODUCTION

Periodic motion and mechanical waves are foundational topics in undergraduate mechanics, with broad applications ranging from wind and ocean waves to seismic activity and engineered vibration systems. In both natural and engineered environments, controlled oscillation is centrally applied to understanding dynamic responses of structures and materials. Shake tables are a primary experimental tool used to simulate mechanical waves and assess the dynamic performance of physical systems under controlled vibration loads.[1,2]

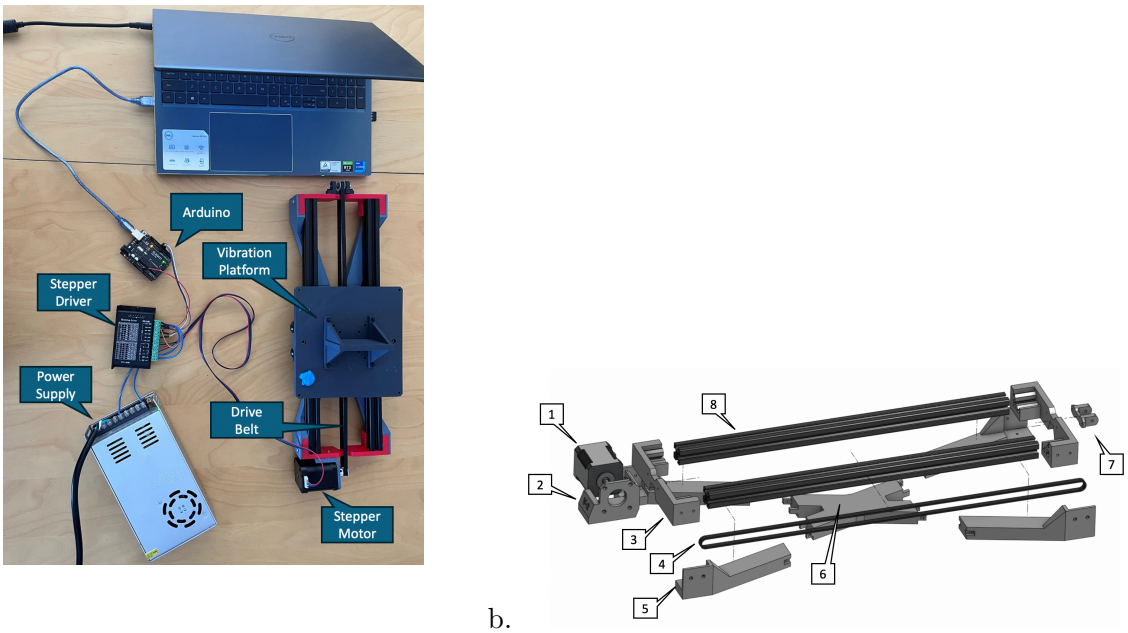
In engineering practice, shaker tables are widely employed in environmental and structural testing to validate that designs can endure intense vibrations, whether induced by earthquakes, machinery, or transportation systems.[3,4] While numerical simulation tools such as ANSYS provide valuable insights into vibration behavior, complex real-world responses, especially of electromechanical systems and intricate assemblies, often require empirical validation through physical testing.[5,6]

Several self-designed shaker tables have been developed to support experimental studies in structural dynamics and vibration testing. Buckley designed a laboratory-scale shaker table for seismic isolator evaluation, emphasizing large lateral displacements and load-bearing capacity; however, the system is relatively large, mechanically complex, and costly.[7] Damcı and Sekerci developed a lower-cost, single-axis Arduino-based shaker table intended for educational purposes; while more compact and affordable than other options, the system is primarily focused on mid-scale earthquake simulation and is still overly heavy and expensive for undergraduate pedagogical purposes.[8] More recently, Damcı et al. presented a bi-axial shaker table driven by stepper motors and controlled via Arduino, capable of reproducing harmonic and earthquake-type motions with good accuracy.[9]

In contrast to these prior implementations, we designed, fabricated, and tested a compact, low-cost single-axis shaker table that can simulate various oscillation modes. Our system uses 3D-printed structural components, a stepper motor actuator, and an Arduino controller to achieve adjustable and easily programmable motion profiles. The result is a shaker table that is not only simple to construct and control, but also well suited for educational and laboratory studies. The design and construction was carried out as a student project while the student was taking undergraduate physics course at calculus level, followed by a semester long independent study course.

II. MATERIALS AND DESIGN

The major components of the shaker table were composed of a stepper motor, a stepper motor driver, an Arduino Uno and a computer, as shown in Fig.1.



a. b.

FIG. 1. a. Top view of the shaker table setup. b. An exploded view of the base of the shaker table. 1: Motor 2: Motor Mount 3: Rail Block 4: Drive Belt 5: Truss 6: Joint 7: Drive Belt Bracket 8: Rail. The CAD document includes the dimensions of each part.[10]

Stepper motors move in steps, which move in constant increments. The stepper motor used for the shaker table is a NEMA-17 (42BYGH60-A-22DH), which has 200 steps per revolution, or 1.8 degrees per step. Microstepping was used to increase the resolution to 400 steps per revolution or 0.9 degrees per step. Maximum acceleration in a belt/pulley/motor is a function of load, motor torque (rated 0.7 N·m holding torque at 0 speed) and pulley radius (0.006 m). Actual performance (acceleration vs load curves etc.) depends on the motor chosen, particularly torque/speed curve. The lightweight construction of this shaker table can become limiting when forces become large (high load at high frequency and amplitude); when pushing the envelope the shaker table must be securely clamped to prevent unwanted motion. Maximum acceleration and load can be increased by using a more powerful motor,

although in that case it may be advisable to modify the design to increase robustness such as by using thicker structural members or fiber-reinforced printer filament, a thicker/stronger drive belt, making provision for bolting to a table.

A TB6600, a common stepper motor driver, was used to control the NEMA-17. This industrial grade stepper motor driver is commonly used for Computer Numerical Control machines and other automated industrial machines, and is inexpensive ($9.98 out of a total materials cost of $186.38 in materials, Bill of Materials provided on Github)[11] while having features such as selectable current and easy micro-stepping. After repeated trials, a 24V, 15A power supply was chosen to power the stepper motor driver, providing a larger working range for the stepper motor.

Using a stepper motor controller and the AccelStepper software library, an Arduino Uno R3 was used to program motion modes.[12] All software was written in the Arduino IDE. Pre-defined motion profiles were implemented as functions that accept user-defined parameters. The code *ShakerTableV2Final-TB6600.ino*, was thoroughly commented and documents the primary software functions.[11]

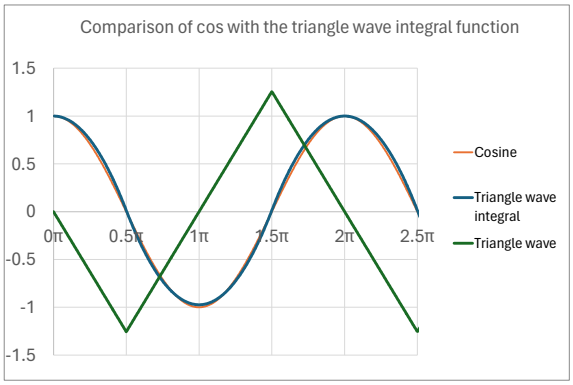
Compared with other shaker table designs reported in the literature,[4789] the present design is less expensive, lighter (approximately 2.7 kg), and mechanically simpler. It is constructed entirely from off-the-shelf and 3D-printed components, making it particularly well suited for undergraduate laboratory and project work. Its low weight and compact size also allow it to be transported easily for classroom demonstrations. For comparison, the simplest alternative design identified in the literature,[8] costs approximately $770, weighs 50 kg,[9] and requires the machining of custom components. The capabilities of the present shaker table are correspondingly more limited. For example, Damci's design has a reported payload capacity of 200 kg,[8] whereas the payload capacity of the present design has only been tested up to 0.9 kg and is clearly far below 200 kg. Direct performance comparisons are of limited value because the designs serve different purposes. The present design of this shaker table design is intended for physics pedagogy.

This paper demonstrates three types of oscillations including sinusoidal, trapezoidal and triangular waves. A sinusoidal wave is the most fundamental waveform and all others can be constructed from a set of sinusoidal waves such as in the Fourier transform. The other two waveforms are chosen to show the flexibility of the platform and the possibility of programming any desired waveform. This table is intended for, and originated as, a student project

so choosing additional waveforms as appropriate to their research topic will improve their understanding far better than selecting from an existing library of earthquake waves, for example. First, consider sinusoidal waves caused by the Simple Harmonic Motion (SHM) of a shaker table. Due to the limited computational power of the Arduino Uno and AccelStepper's capacity of calculation complexity, the exact sinusoidal wave is not calculated; rather an approximation is generated by setting and reversing a constant acceleration in a loop. The function which this approximation produces is a piecewise quadratic approximation, equivalent to integral of a triangle wave.

Given that the magnitude of acceleration is constant and the direction reverses periodically, the acceleration $x''(t)$ has the form of a square wave. As the integral of a constant is a linear function, the integral of a square wave with respect to time is a triangle wave. Similarly, as $\int t\,dt = 1/2t^2$, the integral of a triangle wave with respect to time is a wave composed of repeating sections of parabolas. This is analogous to a defining a piecewise function composed of 2nd order Taylor series about 0, π, 2π etc., except that the error of the Taylor series is considerably greater at the edges of each segment ($\pi/2$, $3\pi/2$,$5\pi/2$. etc.) such that the piecewise function would not be continuous.

a.



b.

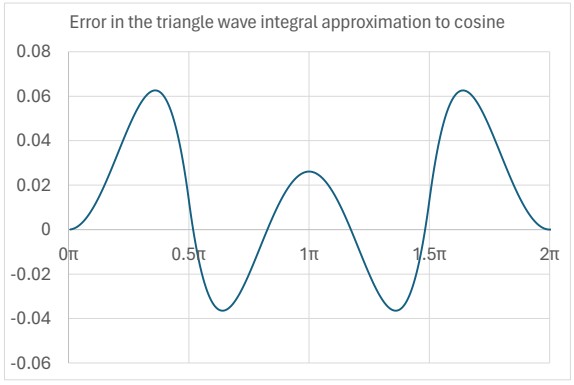


FIG. 2. a. Comparison of the triangle wave and triangle wave integral functions with the cosine function. b: Error of the triangle wave integral function as an approximation to cosine.

In particular, if one considers the triangle wave with a magnitude of $-2\pi/5$, chosen to give the final answer an amplitude of 1 as well as minimize error across the interval of interest, and period of 2π as defined by

$$tri(t) = -4/5|(t - \pi/2)(mod\ 2\pi) - \pi| + 2\pi/5 \tag{1}$$

On the interval t = $[-\pi/2, \pi/2]$,

$$tri(t) = -4/5t \tag{2}$$

$$\int tri(t)\,dt = -2/5t^2 + c \tag{3}$$

Plotting $\int tri(t)\,dt$ where c = 1 (Fig. 2a, blue line) on top of $cos(t)$ (Fig. 2a, orange line) shows the close correspondence of the two functions. A plot of the error curve $\int tri(t)\,dt - cos(t)$ is given in Fig. 2b. By numerical spreadsheet calculation it may be shown that the maximum error of the approximation is $\approxeq 0.06$ and the mean square error is $\approxeq 0.001$.

For the code implementation, one begins by establishing the relationship between the table motion and the motor parameters. The user input amplitude A is converted into motor steps as

$$A_{steps} = A\frac{n}{d}, \tag{4}$$

where n is the number of steps per revolution of the step motor (400) and d is the linear distance traveled per motor revolution (0.04241 m). For sinusoidal motion, the constant acceleration is set as its maximum value. According to the SHM definition, the maximum acceleration is determined from the given frequency f and the maximum displacement, amplitude A.

$$a_{max} = (2\pi f)^2 x|_{max} = (2\pi f)^2 A \tag{5}$$

In this case, $a_{max} = (2\pi f)^2 A_{steps}$. Since the function is only approximately a sinusoidal wave, this formula is only approximately correct; it results in a frequency that is high by a factor of $\sqrt{5}/2 \approx 1.118$. When the target position at $\pm A$ is reached, the table's movement is repeated in the opposite direction. As a result, the motion of the shaker table approximately generates a sinusoidal waveform.

The mechanical parts of the shaker table were 3D printed with the design goals of being rigid, compact and lightweight. As shown in Fig.1, the shaker table/vibration platform rides on two roller bearing plates on standard 0.020 m×0.020 m aluminum extrusion. This provides a smooth rolling, sturdy surface to mount test pieces. The belt also attaches to the underside of the table, providing the locomotion for the table. Each extrusion rail is made parallel to each other with the rail blocks on each end. The extrusions are press-fit into the blocks, and the blocks are bolted onto the truss. This allows easy maintenance on the truss if needed, as by removing two screws the whole frame can be removed from the trusses.

The trusses provide the rigidity for the shaker table. The trusses are slightly undersized and put the whole shaker table frame under slight tension when assembled, which improves rigidity. There are also rubber feet inserted into each corner of the table, which keep it from moving around while running.

III. RESULTS AND TEACHING PRACTICE

A. Results and discussion

The shaker table motion was tested using an ultrasonic sensor from Pasco positioned in front of a flat object mounted on the table.[13] Pasco Capstone software was used to record the motion data.[14] The data acqisition frequency can be adjusted up to 250 Hz in Capstone with the testing range 0.15 to 4.0 m in the front of the sensor. While collecting position values, Capstone can also produce the velocity and acceleration graphs if needed.

Fig. 3 presents the position data of three motion modes collected in Capstone: sinusoidal, trapezoidal and triangular wave forms. The sinusoidal motion generated by the shaker table is clearly observed in Fig. 3a, where curve fitting performed using Capstone yields low root-mean-square error (RMSE), indicating good agreement with the expected waveform. At frequencies at or below 1.0 Hz, a notable observation emerges when a random sample interval between 0 and 1.0 s is examined in greater detail. "Steps" appear in the measured position data, each corresponding to an individual step of the stepper motor. They are not visually perceptible in the physical motion of the shaker table, but can be directly felt by placing one's finger on the pulley while the drive belt is disconnected at low speed ($\sim$ 0.1 Hz).

Fig. 3b shows the position profile for a trapezoid waveform. In this case the parameters (velocity of 0.1 m/s, amplitude of 0.002 m, and delay of 0.5 s) were chosen to resemble a square wave. A true square wave cannot physically exist, as it would consist of repeated instantaneous teleportation. The trapezoid wave still requires very rapid acceleration, to approximate the instantaneous acceleration of an ideal trapezoid wave. This high acceleration is hypothesized to be the cause of the slow drift of the platform towards the non-motor end of the apparatus observed when the trapezoid wave is used; stepper motors are known to skip steps when high accelerations are demanded, although it is unclear why this occurs

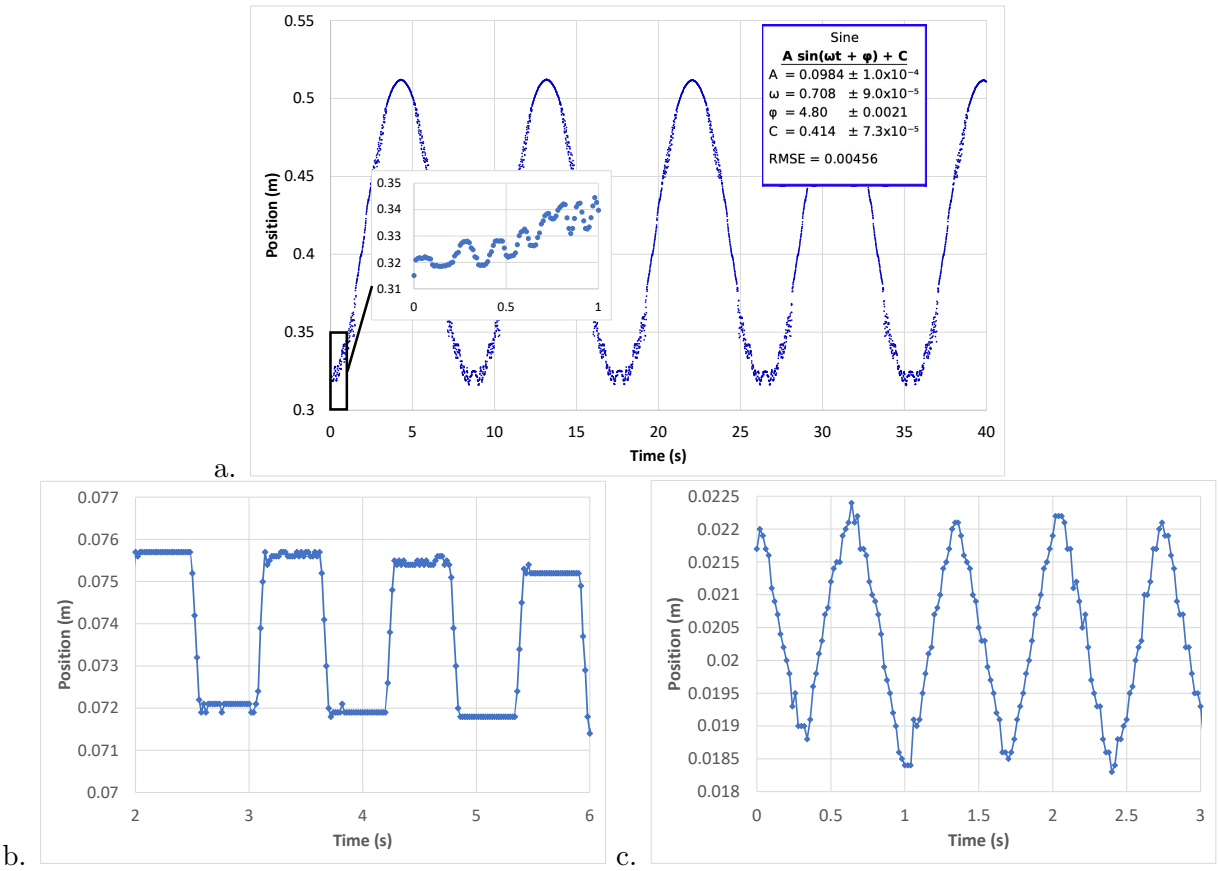
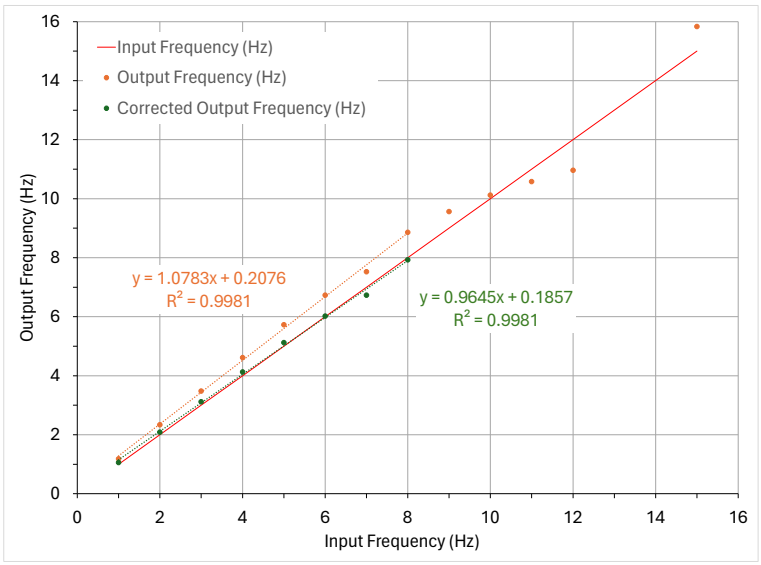


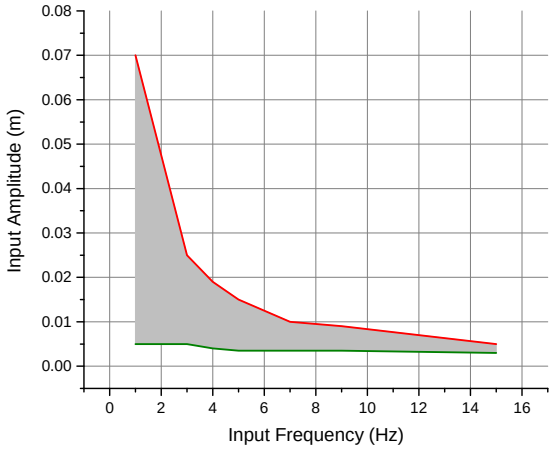
FIG. 3. Sample of sinusoidal waves (a), trapezoid waves (b) and triangle waves (c) produced by the shaker table. In Fig. 3a, angular frequency is calculated as $\omega = 2\pi f$ and φ is phase constant. Fig. 3a inset is the close-up view highlighting motor step-induced vibration visible at low frequencies.

unidirectionally. The drift was noted in multiple trials, does not change when the table is reoriented and becomes visually observable as runtime is extended. A triangle wave can be produced by setting the delay (which sets the time between legs of the trapezoid) to zero (an example as shown in Fig. 3c).

Fig. 4a compares input frequencies with the frequencies achieved using input amplitudes as, 0.010 m at 6 Hz or less; 0.008 m between 7 and 12 Hz, and 0.005 m at 15 Hz. The achieved frequency is calculated from ω obtained through the curve fit in Fig. 3a. The achieved frequency data have been run in triplicate and the standard deviations of all frequency points are under 0.01%. The shaker table oscillates with the discrepancy increasing at frequencies at or below 8 Hz. Over the range up to 15 Hz, the error remains relatively small (between 1.2% and 8.6%).



a.



b.

FIG. 4. a. Output (achieved) frequencies and corrected output frequencies in comparison with input frequencies. b. Diagram of the frequency-amplitude operating window, constructed from input frequencies and amplitudes where the shaker table performed within a 10.0% error margin of calibrated input frequency. The testing data is available on the project Github page.[11]

At frequencies at or below 8 Hz, the primary error source is the waveform approximation discussed below Eq. (5), which can be corrected accordingly. As shown in Fig. 4a this systematic difference is compensated through calibration by applying a correction factor of 1.118. Applying this correction factor results in improved frequency accuracy as demonstrated by the change in regression slope from 1.08 to 0.96. In practice, it is recommended that the users apply calibration for frequencies at or below 8 Hz to achieve desired output

frequencies by dividing the target frequency by 1.118 before programming the shaker table. For example, an input frequency of 3/1.118 Hz should be used to achieve 3 Hz vibration out from the shaker table.

A detailed study of the frequency-amplitude operating envelope for sinusoidal waves from 1 to 15 Hz was undertaken. The envelope was defined as the region where the achieved frequency was within 10% of the calibrated input frequency. The edges of the envelope were located using a hybrid of binary search and successive estimation. The 10% frequency error cutoff is chosen to avoid the onset of substantial waveform distortion. These distorted waveforms approach an approximate triangle wave when input amplitude values are larger than the upper edge of the envelope. By analyzing the data points on the upper boundary of the envelope, a constant proportional to power is obtained,

$$f_{obs}^2 \cdot A_{obs}^2 \approx 0.005 \pm 0.0004 \; Hz^2 \cdot m^2 \tag{6}$$

strongly supporting the hypothesis that performance is limited by motor power output.

A secondary limitation was observed at 15 Hz, where only 0.002 m above the operating region the drive belt pulled loose from its anchors. Within the operating envelope at 15 Hz, the drive belt exhibits substantial vertical oscillations, and suggesting that the apparatus may be approaching a mechanical resonance. Alternatively, the apparatus may be approaching the limits of its materials of construction. The lower edge of the operating region exhibits only a slight downward slope with increasing frequency, resulting in a narrowing of the operating envelope at high frequency. At 15 Hz, the operating range is only from 0.0030 to 0.0050 m input amplitude or 0.0040 to 0.0046 m achieved amplitude. Thus, 15 Hz can be considered the practical upper frequency limit. Users should bear in mind that these measurements were taken on the table without any load shown in Fig. 1; adding additional mass to the table will increase the power required and alter the envelope accordingly to satisfy the constant power relationship.

Any belt-driven system may experience backlash due to the finite elasticity of all materials. The drive belt chosen for this project is a GT2 fiberglass-reinforced 3d-printer belt designed to minimize backlash-related positioning error. Backlash is best minimized by staying within the operating region to avoid subjecting the structure to high loads. Choosing an appropriate belt tension, according to the belt and motor manufacturer specifications, may also help minimize backlash. Backlash affects the position of the table and thus the

waveform and amplitude, rather than frequency.

B. Teaching practice

This low-cost and accessible shaker table is designed specifically for undergraduate physics and engineering students, serving as an instructional tool to support learning wave properties, vibration analysis, and mechanical resonance. Compared to other oscillation motion sources, the shaker table can be easily and precisely controlled, providing various programmable periodic motion modes. In order to help students achieve learning goals regarding periodic motion, the following didactical framework is organized into three progressive levels of complexity.

Level One: At this introductory level, students focus on the description and analysis of oscillations. The activities are designed to teach waveform characterization and foundational kinematics based on the Backward Design curriculum development approach.[15] One of the learning goals of the chapter on periodic motion is to describe oscillations in terms of amplitude, period, frequency, and angular frequency. For students' early learning in the chapter on motion along a straight line, one of the learning goals is to interpret graphs of position versus time, velocity versus time, and acceleration versus time.

Kinematic Parameterization: By using ultrasonic motion sensors and data collection and analysis software, such as Pasco Capstone, the class records and presents similar real-time position diagrams as shown in Fig. 3. Students read the diagrams and describe these oscillations quantitatively using variables such as amplitude, period, frequency, and angular frequency. This activity can serve either as reinforcement practice following instruction on periodic motion or as a brief in-class formative assessment.

Predictive Analysis through Calculus: Students are tasked with applying the definitions of velocity and acceleration (as derivatives of position) to predict the shapes of $v(t)$ and $a(t)$ graphs based on given position diagrams. This activity reinforces the relationship between position, velocity, and acceleration learned in early mechanics. After predicting the outcomes, students have a peer discussion about their work then compare it with actual data collection and software analysis.

The new add-on to the above two activities is using the shaker table for the wave diagrams, rather than just using diagrams as would have been done previously. As a result, the allotted

class time isn't substantially increased. The suggested class time of each activity is 15-20 mins. The other option is to provide students the setup to explore with guided questions as an assignment.

Waveform Validation: After students characterize different motion modes, they compare experimental data with the input data of theoretical models to evaluate how closely the physical system reproduces ideal waveforms. Using a model-based learning approach, they move beyond "ideal" physics to confront the complexities of real-world data. Through curve fitting in Capstone and calculation of RMSE, students quantitatively assess the fidelity of the system, including application of linear regression to calibrate the device for improved accuracy.

Physical Constraints Exercise: This component encourages critical thinking about the intersection of physics and engineering. The instructor can guide a discussion on why a true square wave is physically impossible, as it would require infinite acceleration. Students can also be guided to observe the "drift" of the platform or the motor skipping steps when high accelerations are demanded for trapezoidal waves. As a learning outcome, students develop an appreciation for the inherent constraints of electromechanical systems and recognize that physical systems can only approximate mathematical ideals at a certain level.

Level Two: Hands-On Assembly and Computational Modeling. The second level shifts toward a hands-on construction and programming approach, bridging the gap between theoretical physics and applied engineering. Unlike the first level designed as a structured, whole-class activity, the second level is better suited for individual or small-group course projects due to the increased difficulty, complexity, and time commitment.

Students build the system themselves using 3D-printed structural components, a NEMA-17 stepper motor, and an Arduino Uno controller. Using the sinusoidal waveform as an example, they translate the theoretical framework of SHM into a tangible, programmable system. By applying the defining characteristics of SHM, students derive Eq. (5) and implement mathematical approximations in software. Through interaction with the AccelStepper library, students learn how continuous physical phenomena are simulated using discrete steps. They examine how the Arduino produces sinusoidal motion via piece-wise quadratic approximations.

The second level emphasizes project-based and experiential learning, fostering problem-solving and iterative design thinking. Students make informed design decisions, debug code,

and refine mechanical performance through cycles of testing and revision. The activities and assessments at this level target higher-order learning outcomes. Students evaluate hardware limitations, emphasize error analysis and calibration, and apply linear regression to address systematic discrepancies. Through modeling, validation, and refinement, they strengthen both their conceptual understanding and competence in quantitative analysis and system design.

Level Three: Structural Dynamics and Advanced Resonance. The final level extends the shaker table's applications to the concept of driving forces and the real-world effects of mechanical waves, with a focus on resonance and structural response. Within the teaching practice, this stage is designed for open-ended investigation and is well-suited as a research project of an undergraduate independent study course for senior physics or engineering majors.

Instructors can deepen students' understanding of driving force and resonance by first eliciting predictions about frequency response, phase relationships, and amplitude amplification, and then testing these predictions experimentally. By systematically varying the driving frequency and measuring system response, students construct resonance curves and identify natural frequencies, strengthening the connection between mathematical models and observable behavior.

The shaker table also supports advanced studies in structural dynamics, enabling students to examine how different oscillation modes affect test structures and to design and evaluate tuned mass dampers under seismic-like excitation. Engagement is enhanced through case-based learning that connects laboratory findings to well-known resonance examples, such as the Tacoma Narrows Bridge collapse and the tuned mass damper in Taipei 101. Students may observe resonance by varying the vibration frequency of the table, thus verifying the natural frequency of a 3d-printed structure. Additionally, students may observe the reduction in vibration when a tuned mass damper is added. At this level, students transition from passively observing resonance phenomena to empirically investigating and modeling them in a laboratory setting, thereby deepening both conceptual understanding and practical engineering competence.

IV. CONCLUSION

A shaker table has been developed from readily available components and custom 3D printed parts, utilizing an Arduino Uno as a programmable wave function generator in combination with a stepper motor and motor controller. The resulting system is capable of operating at frequencies up to 15 Hz. Wider operating ranges could be achieved by using more powerful motors if required. The system can produce sinusoidal waves, as well as a flexible trapezoid wave which can generate any waveform from triangle waves to near-square waves. Due to the programmable nature of the Arduino platform, a wide variety of other wave forms could be implemented as needed. Beyond its technical versatility, this shaker table provides a compact and cost-effective setup for instruction of essential physical concepts of periodic motion in undergraduate physics curriculum. By implementing the three-level pedagogical framework described in this study, instructors can effectively guide students from foundational kinematic analysis to hands-on system assembly and, ultimately, to advanced independent research in structural dynamics and resonance.

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